Department-Scoped RAG Flow — Qwen selects the departments at query time

DEPARTMENT CATALOG

Departments collection
_id · name · description
keywords · example questions

department options

catalog for routing

1 · INGESTION (user uploads PDF and picks departments)

User uploads PDF
+ selects Department(s) manually

Save PDF to server storage

Create Document record
Documents: filename · departmentIds · status

Extract text
pdf-parse

Split into overlapping chunks
~800 tokens · 15% overlap · keep page no.

Generate chunk embeddings
Ollama → nomic-embed-text

Chunks collection
documentId · departmentIds
page · text · embedding

stored chunks

2 · QUERY (question only — no department selection)

User sends question

in parallel

Load department list
name + description + keywords

Generate question embedding
Ollama → nomic-embed-text

ROUTER call → Qwen
question + department list
returns strict JSON, no prose

Valid JSON and ≥ 1 department
above confidence?

yes

no

Use routed departmentIds

FALLBACK: search all departments
flag route as low confidence

document → department

Find documents where
departmentIds $in routedIds

Load chunks by documentIds

Cosine similarity
question embedding vs chunk embeddings

Sort by score → keep top 5
above similarity threshold

Any chunk above
threshold?

yes

no

Build context
chunk text + document name + page

Reply: not enough information
offer to widen search to all departments

ANSWER call → Qwen
context + question + cite sources

Return answer + citations
+ departments used (user can correct)